

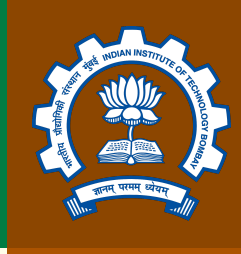
Benchmarking and Performance Analysis of LLM Inferencing Systems

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Contents



1 The Core Trade-off

2 Chunked Hash Tree

3 RL Problem Formulation

4 Policies

The Core Trade-off



- Larger Batches $\rightarrow$ Higher GPU Utilization
- Larger Shared Prefix $\rightarrow$ Better Temporal and Spatial Locality
- Need to ensure the prefix is the same across **all** the requests

Changes to Chunked Hash Tree



- Previously, we used to find the cost of adding a request to the working set (in terms of the newer chunks)
- Now we need information regarding the shared prefix chunks of all the active requests, along with changes to this upon adding a new request
- **Shared Prefix Chain:** A vector of hash blocks which is identical across all the active requests
- On activation: We iterate from back, and see if adding the request removes any blocks?
- On finishing: We iterate over the remaining blocks of the finished request, and check if any other blocks are shared across all the other active requests, and can be added to this chain.

State



- **B:** current batch size
- $\mathbf{C_B}$: number of chunks shared before adding a new request
- $\mathbf{C_A}$: number of chunks which will remain shared after adding the new (best) request
- $\mathbf{R_W}$: number of requests waiting at the end of C_A

State

$s_t \in (B, C_B, C_A, R_W)$

Action



- **ADD:** pop and activate the best request from the chunked hash tree
- **SKIP:** break the batch construction loop, dispatch the batch to GPU

Action

$a_t \in (\texttt{ADD}, \texttt{SKIP})$

Reward



- Negative decode wall step time of the previous batch
- Delivered once per batch

Reward

$$r_t = -t_{\texttt{exec}}^{(t-1)}$$

Policy 1: Q-Learning Scheduler



State Discretization

- Discretize continuous state into buckets:

$$\tilde{s}_t = (\tilde{B}, \widetilde{C_B - C_A}, \tilde{R}_W)$$

- Buckets defined for:
  - Batch size B
  - Chunk difference $C_B - C_A$
  - Requests waiting R_W

Q-Function

$$Q(s_t, a_t)$$

Stored as lookup table:

$$Q : (\tilde{s}, a) \to \mathbb{R}$$

Q-Learning: Action Selection



Epsilon-Greedy Policy

$$a_t = \begin{cases} \text{random action} & \text{with probability } \epsilon \\ \arg\max_a Q(s_t, a) & \text{otherwise} \end{cases}$$

Exploration Decay (Linear)

$$\epsilon_t = \begin{cases} \epsilon_{\text{start}} - (\epsilon_{\text{start}} - \epsilon_{\text{end}}) \frac{t}{T_{\text{decay}}} & t < T_{\text{decay}} \\ \epsilon_{\text{end}} & \text{otherwise} \end{cases}$$

Q-Learning: Update Rule



Temporal Difference Target

$$y_t = r_t$$

Q-Update

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left(y_t - Q(s_t, a_t) \right)$$

Objective:

$$\max_{\pi} \ \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right]$$

Policy 2: Contextual Bandit (UCB)



Simplified State

$$\hat{s}_t = \text{bin}(B, C_B - C_A, R_W)$$

Statistics Maintained

$$N(\hat{s}, a) \quad \text{(action count)}$$

$$\bar{r}(\hat{s}, a) = \frac{\sum r}{N(\hat{s}, a)}$$

Upper Confidence Bound (UCB)

$$\text{UCB}(\hat{s}, a) = \bar{r}(\hat{s}, a) + c\sqrt{\frac{\log T}{N(\hat{s}, a)}}$$

Policy

$$a_t = \arg\max_a \text{UCB}(\hat{s}_t, a)$$

Policy 3: Heuristic Scheduler



Chunk Loss

$$\Delta C = C_B - C_A$$

Decision Rules

1. If $B = 0 \rightarrow$ ADD
2. If $\Delta C \leq 0 \rightarrow$ ADD
3. If $B < B_{\min}$:
   ADD if $\Delta C \leq C_{\max} + 2$
4. If $B \geq B_{\min}$:
   ADD if $\Delta C \leq C_{\max}$
5. If many requests waiting:
   ADD if $R_W \geq 10$ and $\Delta C \leq C_{\max} + 1$
6. Otherwise $\rightarrow$ SKIP



Thank You